Recall:

Linear binary code with length n and dimension k
$\Longleftrightarrow$ k-dimensional subspace of $\mathbb{F}_2^n$

Basis for a linear code C is a subset $B \subseteq C$ such that
- $\langle B \rangle = C$
- B is a linearly independent set

A generating matrix for a linear code C of length n and dimension k is a $k \times n$ matrix whose rows form a basis for C.

There are many different generating matrices for a code C, but only one in RREF.

A linear code generated by a generating matrix of the form $G = (I_k \mid X)$ is called systematic.

Messages $m \in \mathbb{F}_2^k$ are encoded as mG.

<u>Exercise</u> Consider the linear code $C = \langle S \rangle$ for
$S = \{1100111,\ 1101010,\ 1011001,\ 0111110\}$.

(a) Determine the generating matrix in rref for C.

(b) What is the length n and dimension k for C.

(c) Is the code C systematic?

(d) Assuming that your matrix from part (a) was used for the encoding process, determine the message word corresponding to codeword 0111110.

4.6. Dual Codes

We shall now see how to derive a new code from a given linear code C, using the orthogonal complement. A generating matrix for this new code is the matrix $\mathbf{H}$, which is very useful for error detection and correction in the original code C.

Recall that vectors v, $w \in \mathbb{F}_2^n$ are *orthogonal* if $v \cdot w = 0 \in \mathbb{F}_2$. We write $v \perp w$.

If $\emptyset \neq S \subseteq \mathbb{F}_2^n$, the set of vectors orthogonal to every vector in S is denoted $S^{\perp}$.

$$S^{\perp} = \{v \in \mathbb{F}_2^n \mid v \cdot s = 0 \text{ for all } s \in S\}.$$

Exercise Let $S = \{110, 011\}$. Determine $S^{\perp}$.

4.6.2 Lemma For any non-empty set $S \subseteq \mathbb{F}_2^n$, the set $S^{\perp}$ is a vector space (linear code).

4.6.3 Lemma If $S \subseteq \mathbb{F}_2^n$, then $\langle S \rangle^{\perp} = S^{\perp}$.

4.6.4 Lemma Suppose U, V are subspaces of $\mathbb{F}_2^n$, S is a basis for U, T is a basis for V and $s \cdot t = 0$ for all $s \in S$ and $t \in T$. Then $u \cdot v = 0$ for all $u \in U$ and all $v \in V$.

Proof Let $u \in U$ and $v \in V$. Then $u = s_1 + \cdots + s_p$ and $v = t_1 + \cdots + t_q$ where $s_1, \ldots, s_p \in S$ and $t_1, \ldots, t_q \in T$. Thus, $u \cdot v = (s_1 + \cdots + s_p) \cdot (t_1 + \cdots + t_q)$. Since each term in the expansion of $(s_1 + \cdots + s_p) \cdot (t_1 + \cdots + t_q)$ is 0 we have $u \cdot v = 0$. □

4.6.5 Definition If $C \subseteq \mathbb{F}_2^n$ is a code (linear or not), the set $C^{\perp}$ is called the *dual code* of C.

Suppose C is a linear code generated by $S \subseteq \mathbb{F}_2^n$. That is, $C = \langle S \rangle$. Then by Lemma 4.6.3 we have $C^{\perp} = \langle S \rangle^{\perp} = S^{\perp}$. Thus,

$$C = \langle S \rangle \implies C^{\perp} = S^{\perp}. \tag{4.6.1}$$

4.6.6 Theorem Let $C \subseteq \mathbb{F}_2^n$ be a linear code. Then $\dim C + \dim C^\perp = n$.

Proof Let $\dim C = k$. Let $S = \{c_1, \ldots, c_k\}$ be a basis for C. By Equation 4.6.1

$$x \in C^\perp \iff c_j \cdot x = 0 \qquad \text{for } j = 1, \ldots, k.$$

Put the vectors c_j in the rows of a $k \times n$ matrix $\mathbf{A}$. Since $\mathbf{A}$ has linearly independent rows, rank $(\mathbf{A}) = \dim C = k$. Now, $x \in C^\perp \iff \mathbf{A}\,x^\mathsf{T} = \mathbf{0}$. So $\dim C^\perp = \dim \mathrm{NS}(\mathbf{A})$. Thus, from Theorem 4.1.5, $n = \text{rank}\,(\mathbf{A}) + \dim \mathrm{NS}(\mathbf{A}) = \dim C + \dim C^\perp$. □

$$\begin{array}{cc} A & x^T \\ \begin{pmatrix} \text{—}\; c_1 \;\text{—} \\ \text{—}\; c_2 \;\text{—} \\ \vdots \\ \text{—}\; c_k \;\text{—} \end{pmatrix} & \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \end{array} = \underset{\sim}{0}$$

<u>Exercise</u> Let C be a $(10, 4)$ - linear code.

How many codewords in C?

How many codewords in $C^\perp$?

4.6.8 Lemma If $V \subseteq \mathbb{F}_2^n$ is a subspace then $(V^\perp)^\perp = V$.

<u>Pf</u> $\forall x \in V$, $y \in V^\perp$ $\quad y \cdot x = 0$ and $x \cdot y = 0$ so $x \in (V^\perp)^\perp$

$\therefore V \subseteq (V^\perp)^\perp$

$$\left.\begin{array}{l} \dim V + \dim V^\perp = n \\ \dim V^\perp + \dim (V^\perp)^\perp = n \end{array}\right\} \text{so } \dim (V^\perp)^\perp = \dim V$$

$$\underset{G}{\begin{pmatrix} \text{---} \; c_1 \; \text{---} \\ \text{---} \; c_2 \; \text{---} \\ \vdots \\ \text{---} \; c_k \; \text{---} \end{pmatrix}} \begin{pmatrix} | & | & & | \\ d_1 & d_2 & \dots & d_{n-k} \\ | & | & & | \end{pmatrix}$$

H

4.6.9 Lemma Let C be a linear (n, k)-code with generating matrix $\mathbf{G}$. Suppose matrix $\mathbf{H}$ is an $n \times (n-k)$ matrix with linearly independent columns. Then the columns of $\mathbf{H}$ form a basis for $C^\perp$ iff $\mathbf{G}\,\mathbf{H} = \mathbf{0}$.

Proof $\implies$ If the columns of $\mathbf{H}$ form a basis for $C^\perp$, then each column h of $\mathbf{H}$ lies in $C^\perp$, and so $c \cdot h = 0$ for all $c \in C$. But this means $\mathbf{G}\,\mathbf{H} = \mathbf{0}$.

$\impliedby$ Suppose $\mathbf{G}\,\mathbf{H} = \mathbf{0}$. Let D be the span of the columns of $\mathbf{H}$. By assumption, the dot product of each row vector of $\mathbf{G}$ with each column vector of $\mathbf{H}$ is 0. Thus $u \cdot v = 0$ for each u in [a basis for] C and v in [a basis for] D. By Lemma 4.6.4, every vector in D is perpendicular to every vector in C, so $D \subseteq C^\perp$. But $\mathbf{H}$ has rank $n-k$, so applying Theorems 4.1.3 and 4.6.6 $\dim D = n - k = n - \dim C = \dim C^\perp$, so $D = C^\perp$. That is, the columns of $\mathbf{H}$ span $C^\perp$, and they are linearly independent, so form a basis. □

4.6.10 Definition Let C be a linear (n, k)-code. An $n \times (n-k)$ matrix $\mathbf{H}$ is called a *parity check matrix* for C if its columns form a basis for $C^\perp$.

Since the same space may have many bases, the matrix $\mathbf{H}$ is not unique.

Equivalently, an $n \times (n-k)$ matrix H with linearly independent columns is a parity check matrix for C if $GH = 0$ where G is a generating matrix for C.

If C is a systematic code with generating matrix $\mathbf{G}$ in standard form, then it is easy to find a parity check matrix $\mathbf{H}$. Let

$$\mathbf{G} = (\mathbf{I}_k \;\; \mathbf{X})$$

where $\mathbf{X}$ is some $k \times (n-k)$ matrix. Let

$$\mathbf{H} = \begin{pmatrix} \mathbf{X} \\ \mathbf{I}_{n-k} \end{pmatrix}.$$

Then: $\mathbf{H}$ is $n \times (n-k)$, and the presence of $\mathbf{I}_{n-k}$ inside $\mathbf{H}$ ensures that the $n-k$ columns of $\mathbf{H}$ are linearly independent. Finally by block multiplication:

$$\mathbf{G}\,\mathbf{H} = (\mathbf{I}_k \;\; \mathbf{X}) \begin{pmatrix} \mathbf{X} \\ \mathbf{I}_{n-k} \end{pmatrix} = \mathbf{I}_k\,\mathbf{X} + \mathbf{X}\,\mathbf{I}_{n-k} = \mathbf{0}_{k \times (n-k)}.$$

By Lemma 4.6.9, the columns of $\mathbf{H}$ form a basis for $C^\perp$.

Exercise

Let $S = \{10010, 10111, 11001, 11100\}$ and let $C = \langle S \rangle$.

(a) Find the generating matrix in rref for C.

(b) Is C a systematic code?

(c) What are the parameters (n, k, δ) for C?

(d) Find a parity check matrix for C.

(e) How many codewords in $C^{\perp}$?

Even if C is not systematic we can still find a parity check matrix $\mathbf{H}$ as follows:

★ Find a systematic code C' equivalent to C, via some permutation $C \xrightarrow{\sigma} C'$.

★ If $\mathbf{G}$ and $\mathbf{G}'$ are generating matrices for C and C' respectively, then $\mathbf{G} \xrightarrow{\sigma} \mathbf{G}'$ by permuting columns.

★ Since $\mathbf{G}'$ is systematic, find corresponding $\mathbf{H}'$ as above. Thus $\mathbf{H}'$ is a parity check matrix for C'.

★ Finally apply the <u>inverse permutation</u> σ^{-1} to the <u>rows</u> of $\mathbf{H}'$ to obtain $\mathbf{H}$.

4.6.15 Example If $S = \{101010,\ 010101,\ 111111,\ 000111,\ 101100\}$, find a basis for $C = \langle S \rangle$, and a basis for $C^{\perp}$.

Solution:

$$\mathbf{A} = \begin{pmatrix} 1&0&1&0&1&0\\ 0&1&0&1&0&1\\ 1&1&1&1&1&1\\ 0&0&0&1&1&1\\ 1&0&1&1&0&0 \end{pmatrix} \to \begin{pmatrix} 1&0&1&0&1&0\\ 0&1&0&1&0&1\\ 0&1&0&1&0&1\\ 0&0&0&1&1&1\\ 0&0&0&1&1&0 \end{pmatrix} \to \begin{pmatrix} 1&0&1&0&1&0\\ 0&1&0&1&0&1\\ 0&0&0&0&0&0\\ 0&0&0&1&1&1\\ 0&0&0&0&0&1 \end{pmatrix}$$

$$\to \begin{pmatrix} 1&0&1&0&1&0\\ 0&1&0&0&1&0\\ 0&0&0&0&0&0\\ 0&0&0&1&1&1\\ 0&0&0&0&0&1 \end{pmatrix} \to \begin{pmatrix} 1&0&1&0&1&0\\ 0&1&0&0&1&0\\ 0&0&0&0&0&0\\ 0&0&0&1&1&0\\ 0&0&0&0&0&1 \end{pmatrix} \to \begin{pmatrix} 1&0&1&0&1&0\\ 0&1&0&0&1&0\\ 0&0&0&1&1&0\\ 0&0&0&0&0&1\\ 0&0&0&0&0&0 \end{pmatrix}.$$

So $\mathbf{G} = \begin{pmatrix} 1&0&1&0&1&0\\ 0&1&0&0&1&0\\ 0&0&0&1&1&0\\ 0&0&0&0&0&1 \end{pmatrix}$. A basis for C is $\{101010,\ 010010,\ 000110,\ 000001\}$, so $\dim C = 4$ and $|C| = 2^4 = 16$.

$$\mathbf{G}' = \left(\begin{array}{cccc|cc} 1&0&0&0&1&1\\ 0&1&0&0&1&0\\ 0&0&1&0&1&0\\ 0&0&0&1&0&0 \end{array}\right) = (\ \mathbf{I}_4 \quad \mathbf{X}\ ), \qquad \mathbf{X} = \begin{pmatrix} 1&1\\ 1&0\\ 1&0\\ 0&0 \end{pmatrix}.$$

Permutation $\sigma\colon 3 \mapsto 6, \quad 6 \mapsto 4, \quad 4 \mapsto 3$ also written $\sigma = (3\ 6\ 4)$.

So $\sigma^{-1}\colon 3 \mapsto 4, \quad 4 \mapsto 6, \quad 6 \mapsto 3, \quad \sigma^{-1} = (3\ 4\ 6)$, so

$$\mathbf{H}' = \left(\begin{array}{ccc} 1&1&\\ 1&0&\\ 1&0&[3]\\ 0&0&[4]\\ \hline 1&0&\\ 0&1&[6] \end{array}\right) \qquad \mathbf{H} = \begin{pmatrix} 1&1&\\ 1&0&\\ 0&1&[6\mapsto 3]\\ 1&0&[3\mapsto 4]\\ 1&0&\\ 0&0&[4\mapsto 6] \end{pmatrix}.$$

A basis for $C^{\perp}$ is $\{110110,\ 101000\}$, and $\dim C^{\perp} = 2$, so $|C^{\perp}| = 2^2 = 4$.

4.7. Parity Check Matrices

4.7.1 Theorem A matrix H is a parity check matrix for a linear code C iff H^{T} is a generating matrix for $C^{\perp}$.

Proof By definition, H is a parity check matrix for C iff the columns of H are a basis for $C^{\perp}$, iff the rows of H^{T} are a basis for $C^{\perp}$, iff H^{T} is a generating matrix for $C^{\perp}$. □

So $\mathrm{G}_{C^{\perp}} = \mathrm{H}_C^{\mathsf{T}}$. Replacing C by $C^{\perp}$ and using $(C^{\perp})^{\perp} = C$, we can write $\mathrm{G}_C = \mathrm{H}_{C^{\perp}}^{\mathsf{T}}$, so $\mathrm{H}_{C^{\perp}} = \mathrm{G}_C^{\mathsf{T}}$. This is summarized in the following important diagram:

(4.7.1)	Code	Generating Matrix	Parity Check Matrix

$$
\begin{tikzcd}
\overset{(n,k)}{C} & \overset{k\times n}{\mathrm{G}_C} \arrow[r, "4.6.13"] & \overset{n\times(n-k)}{\mathrm{H}_C} \arrow[dl, "\mathsf{T}"] \\
\overset{(n,n-k)}{C^{\perp}} & \underset{(n-k)\times n}{\mathrm{H}_C^{\mathsf{T}} = \mathrm{G}_{C^{\perp}}} \arrow[r, "4.6.13"'] & \underset{n\times k}{\mathrm{G}_C^{\mathsf{T}} = \mathrm{H}_{C^{\perp}}} \arrow[ul, "\mathsf{T}"]
\end{tikzcd}
$$

Given any one of the generating or parity check matrices for C or $C^{\perp}$, using diagram 4.7.1 we can find the other three matrices. This means: *We can specify C and/or $C^{\perp}$ by giving H or G.*

<u>Exercise</u>: $H_c = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is a parity check matrix for a code C. Find a parity check matrix for the code $C^{\perp}$.

The next theorem shows an important use of the parity check matrix. Recall that by definition of $\mathbf{H}$, the column space of $\mathbf{H}$ is $C^{\perp}$.

4.8.1 Theorem Let C be a linear code with parity check matrix $\mathbf{H}$. Then $C = \{v \in \mathbb{F}_2^n \mid v\mathbf{H} = \mathbf{0}\}$.

Proof Clearly $v\mathbf{H} = \mathbf{0}$ iff v is perpendicular to each column of $\mathbf{H}$ iff v is perpendicular to the column space of $\mathbf{H}$ (see Lemma 4.6.4) iff $v \in (C^{\perp})^{\perp} = C$. □

In particular: *If we receive w and $w\mathbf{H} \neq \mathbf{0}$, then error(s) must have occurred.*

Exercise The matrix $H = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is a parity check matrix for a linear code C.

Which of the following words are codewords of C?

(a) 01110

(b) 10011

(c) 10100

For a linear (n, k)-code, the matrix equation $x\mathbf{H} = \mathbf{0}$, where $x = (x_1, x_2, \ldots, x_n)$, is a shorthand way of writing a set of $n - k$ simultaneous linear equations in the variables $x_1, x_2, \ldots, x_n$. If the jth column of $\mathbf{H}$ has entries of 1 in positions $m_1, m_2, \ldots m_j$, then the bit-sum

$$x_{m_1} + x_{m_2} + \cdots + x_{m_j} = 0.$$

In other words, the binary word x satisfies $x\mathbf{H} = \mathbf{0}$ iff the sums (obtained by ordinary addition) of certain of its bits (as determined by $\mathbf{H}$) are even (have even parity). The $n - k$ linear equations are thus referred to as *parity check equations* of the code, and $\mathbf{H}$ is called the *parity check matrix*.

In summary:

4.8.4 Theorem Matrices $\mathbf{G}$ and $\mathbf{H}$ are generating and parity check matrices respectively for a linear (n, k)-code C iff

- ★ The rows of $\mathbf{G}$ are linearly independent,
- ★ The columns of $\mathbf{H}$ are linearly independent,
- ★ $\mathbf{G}$ is $k \times n$, and $\mathbf{H}$ is $n \times (n-k)$, and
- ★ $\mathbf{G}\,\mathbf{H} = \mathbf{0}$.

Then the rows of $\mathbf{G}$ are a basis for C, and the columns of $\mathbf{H}$ are a basis for $C^{\perp}$. The relationship between $\mathbf{G}$ and $\mathbf{H}$ is summarized in diagram 4.7.1.

(4.7.1)

$$
\begin{tikzcd}
\text{Code} & \text{Generating Matrix} & \text{Parity Check Matrix} \\
\overset{(n,k)}{C} & \overset{k\times n}{\mathbf{G}_C} \arrow[r, bend left, "4.6.13"] & \overset{n\times(n-k)}{\mathbf{H}_C} \arrow[dl, "\mathsf{T}"] \\
\overset{(n,n-k)}{C^{\perp}} & \underset{(n-k)\times n}{\mathbf{H}_C^{\mathsf{T}} = \mathbf{G}_{C^{\perp}}} \arrow[r, "4.6.13"'] & \underset{n\times k}{\mathbf{G}_C^{\mathsf{T}} = \mathbf{H}_{C^{\perp}}} \arrow[ul, "\mathsf{T}"]
\end{tikzcd}
$$